

Conditional and Independent-Event Probabilities

Conditional probability from a two-way table, the multiplication rule, testing independence, and independent trials in a binomial model

Rules used on this sheet:

- $P(A | B) = \frac{P(A \cap B)}{P(B)}$ — in a table, restrict to B 's row or column and divide within it.
- Multiplication rule: $P(A \cap B) = P(B) \cdot P(A | B)$. When A and B are **independent**, $P(A | B) = P(A)$ and this becomes $P(A) \cdot P(B)$.
- A and B are independent exactly when $P(A \cap B) = P(A) \cdot P(B)$.
- Give exact fractions in lowest terms unless a decimal is requested.

Survey table — used by problems 1, 2, 3 and 6. All 200 students at a school were asked whether they ride the bus and whether they play a school sport.

	Plays a sport	No school sport	Total
Rides the bus	48	32	80
Does not ride the bus	72	48	120
Total	120	80	200

1. A student is chosen at random from the 200 surveyed. Given that the student rides the bus, find the probability that the student plays a school sport. Write the conditional probability you are computing in $P(A | B)$ notation before you divide.

Answer: _____

2. Using the same survey, find the probability that a student rides the bus, given that the student plays a school sport. Explain in one sentence why this answer differs from your answer to problem 1 even though both use the 48 students in the same cell.

Answer: _____

3. Still using the survey table, test whether “plays a school sport” and “rides the bus” are independent events.

(a) Find $P(\text{plays a sport})$ for a randomly chosen student.

Answer: _____

(b) Find $P(\text{plays a sport and rides the bus})$.

Answer: _____

(c) Multiply $P(\text{plays a sport}) = 0.6$ by $P(\text{rides the bus}) = 0.4$, compare with your answer to (b), and state whether the two events are independent.

Answer: _____

4. On a production line each memory chip passes inspection with probability 0.98, and the chips are tested independently of one another. Two chips are selected. Find the probability that *both* pass inspection. Give your answer as a decimal.

Answer: _____

5. A drawer holds 12 identical pens, of which 5 write in red. Two pens are drawn at random **without replacement**. Find the probability that both pens write in red. Show the two factors you multiplied, and state the conditional probability that the second factor represents.

Answer: _____

6. Return to the survey table. Given that a student does *not* play a school sport, find the probability that the student rides the bus. State the total you divided by and why that total, not 200, is the correct denominator.

Answer: _____

7. For two events A and B it is known that $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.25$.

(a) Compute the product $P(A) \cdot P(B)$.

Answer: _____

(b) Compute $P(A \mid B)$, then state whether A and B are independent and justify the decision with your two numbers.

Answer: _____

8. A factory gets 60% of its parts from line 1 and 40% from line 2. Of the parts from line 1, 2% are defective; of the parts from line 2, 5% are defective. A part is chosen at random.

(a) Find the probability that the part came from line 1 *and* is defective.

Answer: _____

(b) Find the probability that the part is defective.

Answer: _____

9. A basketball player takes 3 free throws. The throws are independent; write a for the probability of making one throw and b for the probability of missing one, so $a + b = 1$. Expand $(a + b)^3$ completely, then state which term gives the probability of making exactly two of the three throws, and explain why that term carries a coefficient greater than 1.

Answer: _____

10. The same player makes each free throw with probability 0.8, and the three throws are independent.

(a) Find the probability that she makes exactly two of the three throws. Give a decimal.

Answer: _____

(b) Explain in two or three complete sentences why multiplying $0.8 \times 0.8 \times 0.2$ is valid only when the throws are independent, and what would have to change in the calculation if they were not.